Annual Temperature Anomalies Relative To The Pre-Industrial Period

Time Series and Forecasting Report

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1 Introduction

This report explores the historical progression of global surface temperatures through the analysis of annual temperature anomalies. As climate change remains one of the most critical challenges of the 21st century, understanding the underlying patterns of these anomalies is essential for climate modeling and future projections. The following sections detail the data source, its structure, and the temporal scope of the series.

1.1 Data Source and Context

The data for this analysis is retrieved from *Our World in Data* [1], based on the **HadCRUT5** dataset [2]. This dataset is a collaborative effort between the **Met Office Hadley Centre** and the **Climatic Research Unit** at the University of East Anglia. It represents a global grid of historical surface temperature anomalies, combining land-surface air temperatures and sea-surface temperatures to provide a comprehensive view of the Earth's climate evolution.

1.2 Variable Description and Measurement

The dataset includes three primary variables, all measured in degrees Celsius (°C):

- **Global Average Temperature Anomaly:** The primary variable of interest, representing the deviation of the global mean temperature from the specified baseline.
- **Lower Bound:** The lower limit of the 95% confidence interval, accounting for measurement and sampling uncertainties.
- **Upper Bound:** The upper limit of the 95% confidence interval.

The "anomaly" is defined as the difference between the observed temperature and a reference baseline. In this dataset, the baseline period is **1861–1890**, which is widely used as a proxy for the pre-industrial period [1]. A positive anomaly indicates that the observed temperature was warmer than the pre-industrial average, while a negative anomaly indicates it was cooler.

1.3 Temporal Scope and Frequency

The time series consists of **annual observations**, providing a high-level view of long-term climate trends while filtering out short-term seasonal noise. The data spans a total of **175 years**, beginning in **1850** and extending through **2024**. This significant time span allows for the observation of both the stable climate patterns of the late 19th century and the rapid warming trend observed from the mid-20th century to the present day.

2 Time Series Features

2.1 Descriptive Analysis

The exploratory data analysis (EDA) of the global temperature anomaly series (1850–2024) reveals a dataset with 175 annual observations. The central variable is the deviation in degrees Celsius from the 1861–1890 baseline.

- **Summary Statistics:** The series transitions from negative anomalies in the late 19th century (indicating cooler-than-average years) to consistently positive and rising anomalies in the 21st century.
- **Visual Inspection:** The time series plot shows a clear, non-linear upward trend. While the first century of data (1850–1950) exhibits fluctuations around a relatively stable mean, the period from 1970 onwards shows a sharp and sustained acceleration in warming.
- **Volatility:** There is significant year-to-year variability, often attributed to natural cycles like El Niño and La Niña. To isolate the anthropogenic warming signal, it is standard practice to account for this ENSO-related variance, which can contribute to annual global temperature fluctuations [3], though the magnitude of these fluctuations remains small compared to the long-term warming signal.

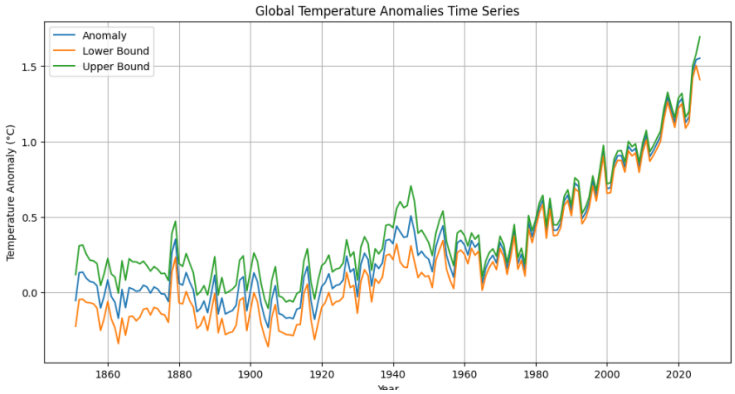


Figure 1: Global Temperature Anomalies (1850–2024). Note the accelerating upward trend in the latter half of the series.

2.2 Time Series Decomposition

Given the annual frequency of the data, the series is modeled using an additive decomposition [4]: $Y_t = T_t + R_t$, where Y_t is the observed anomaly, T_t is the trend, and R_t is the remainder. Since the data is annual, there is no seasonal component ($S_t = 0$), which is confirmed by the absence of rhythmic 12-month cycles in the autocorrelation.

A smoothing technique (such as LOWESS) was applied to extract the **Trend component**, which captures the long-term climatic warming. By isolating this trend, we can observe the underlying movement of the Earth's temperature independent of annual noise.

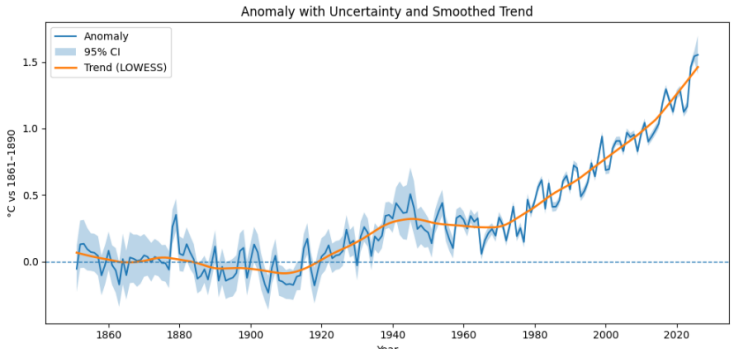


Figure 2: Additive decomposition of the temperature series showing the extracted Trend.

2.3 The Remainder Component

The **Remainder** represents the stochastic part of the series after the trend is removed. Analysis of the remainder reveals:

- **Mean-Zero Property:** The remainder fluctuates around zero, suggesting that the trend extraction was successful in capturing the long-term growth.
- **Persistence:** The remainder is not "pure" white noise; it exhibits brief "runs" or bursts where several consecutive years stay above or below the trend line, suggesting short-term memory in the climate system.
- **Variance:** The amplitude of these deviations generally stays within $\pm 0.1°C$ to $\pm 0.3°C$.

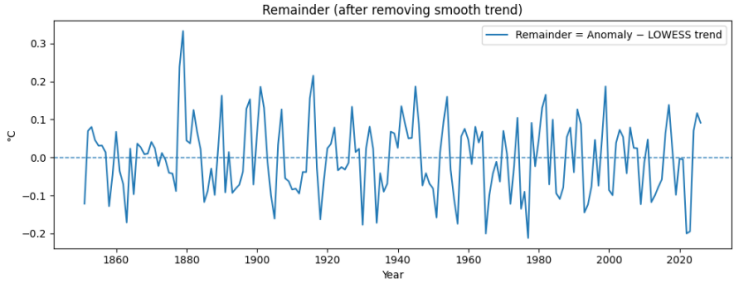


Figure 3: Remainder of the time series.

2.4 Autocorrelation Analysis

To identify the internal structure of the remainder, we analyze the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots.

- **Remainder ACF:** The ACF plot shows a **fast decay to zero** immediately after the initial lags. This confirms that the trend component was effectively removed and that there is no seasonality, as no significant periodic spikes are present. Consequently, the remainder behaves as a **stationary series**. Furthermore, the presence of a significant spike at the first 3 lags followed by an immediate cutoff suggests the potential for a **Moving Average (MA)** component. This indicates that the current year's anomaly is influenced by the "shocks" or random errors of the previous period, which is a classic signature of an MA(3) process.
- **Remainder PACF:** The PACF plot exhibits significant spikes at the first few lags (notably lag 1 and 2) followed by an abrupt cutoff. This pattern is a classic signature of an **Autoregressive (AR) process**, suggesting that the temperature anomaly of a given year retains a direct "memory" of the values from the immediately preceding years.

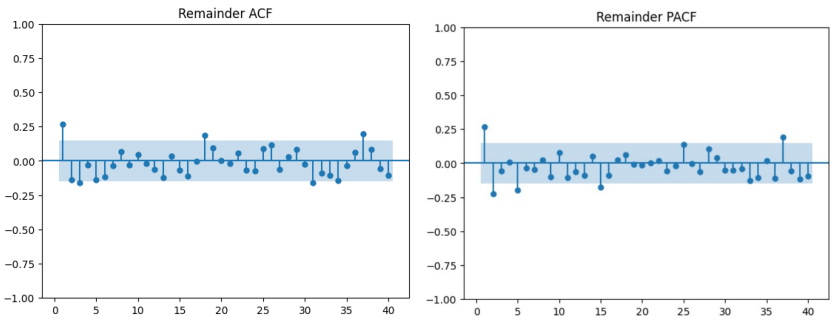


Figure 4: ACF and PACF plots of the remainder component.

3 Training Set and Test Set

Before proceeding to the modelling and forecast sections, the dataset was divided into two groups: a training set, which will be used to find the model that best fits the data, and a test set, to later cross-validate against the forecasted values. The proportion used to apply this division was 80–20, meaning 80% of the data was assigned to the training set, which includes the years from 1850 to 1990 (141 observations), and the remaining 20% to the test set with the years from 1991 to 2025 (35 observations).

4 Modelling

Considering that in our data there is no seasonality, we will focus on models without a seasonal component. To begin, we will apply both AR and MA models, taking into account the information we got from the time series analysis above.
To compare the different models and decide on which to use for the forecast, the metric AIC will be preferred over BIC, considering that AIC aims for the best predictive model and thus finds the best approximating model for future predictions.

4.1 AR Models

For AR models, it is necessary to decide which p order we will use. For that, looking at the PACF plot, it is possible to see a cutoff on lag 2, thus indicating that our AR model should have a p=2. However, we tested AR models with orders 1, 2, and 3 to reach a definitive conclusion (for the results on the models AR(1) and AR(3), please refer to the Appendix).

The model AR(2) got the following results:

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(2, 0, 0)	Log Likelihood	137.417
Date:	Sun, 21 Dec 2025	AIC	-266.834
Time:	23:22:21	BIC	-255.039
Sample:	0	HQIC	-262.041
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0010	0.009	0.106	0.915	-0.017	0.019
ar.L1	0.3221	0.077	4.164	0.000	0.171	0.474
ar.L2	-0.1784	0.087	-2.044	0.041	-0.350	-0.007
sigma2	0.0083	0.001	7.876	0.000	0.006	0.010

Ljung-Box (L1) (Q):	0.06	Jarque-Bera (JB):	0.49
Prob(Q):	0.80	Prob(JB):	0.78
Heteroskedasticity (H):	1.23	Skew:	0.10
Prob(H) (two-sided):	0.49	Kurtosis:	2.78

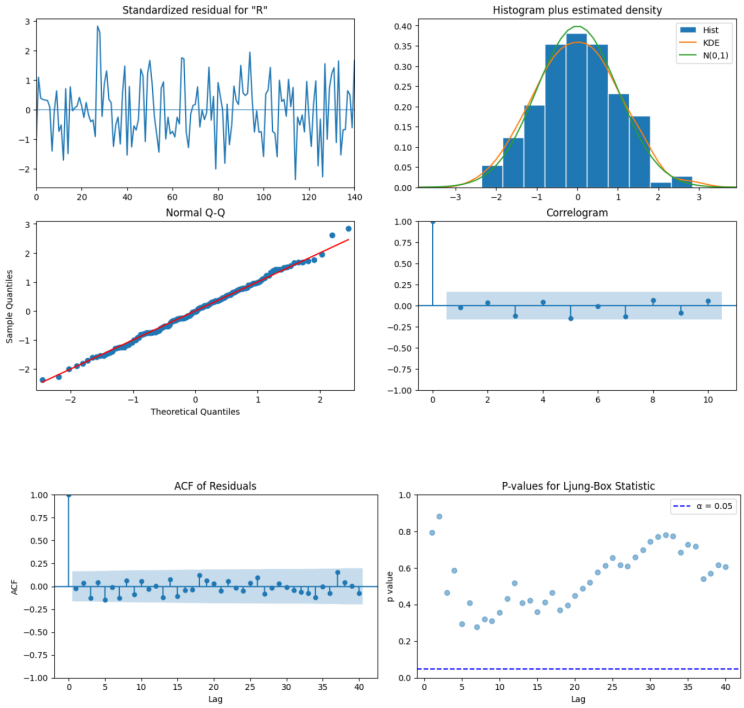


Figure 5: Statistical results for AR(2) and its residuals.

As seen in the images above, the residuals are normally distributed (slightly heavier on the tails), uncorrelated, and with p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant).

When comparing the 3 models calculated, the metrics obtained were:

Model	BIC	AIC	AICc
AR (1)	-1.812261	-1.875001	-1.873758
AR (2)	-1.808786	-1.892439	-1.890353
AR (3)	-1.783844	-1.888410	-1.885257

Table 1: Metrics for AR models

Notice that the metrics in the table vary from the metrics found in the figures, due to a normalization applied where BIC, AIC, and AICc were divided by the length of the set. As mentioned at the beginning of this chapter, the AIC metric will be the preferred one, and, as can be seen in the table above, the AR(2) model has the lowest value, making it the best fit for our dataset.
However, although AR(2) seems to be a good model, according to the parsimony principle [5], a model with fewer parameters that provides a similar or better fit will lead to a more parsimonious model. Moreover, the aim is to find a

global optimum across all ARIMA specifications; thus, MA models will also be calculated.

4.2 MA Models

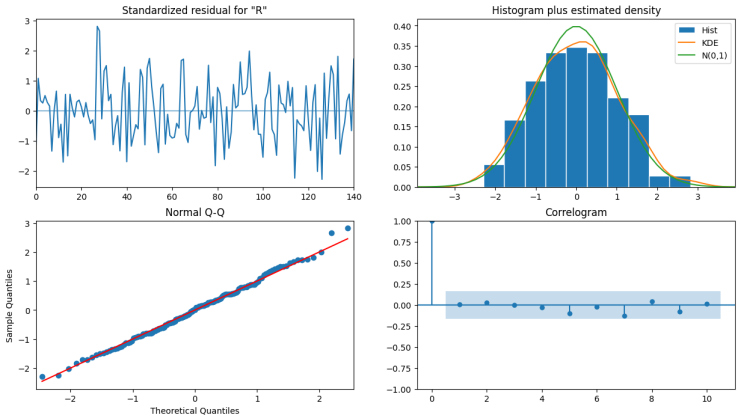
To apply the MA models, it is necessary to first define the q order. Taking into account the ACF plot of our remainder, a cutoff can be seen on lag 1. Even so, lag 3 could be considered the cutoff as well, so in order to decide which model best fits the dataset, similar to AR models, it will be considered orders 1, 2, and 3.
When calculated, the model MA(3) revealed the best results (to check the results for models MA(1) and MA(2), see the Appendix).

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(0, 0, 3)	Log Likelihood	139.109
Date:	Mon, 22 Dec 2025	AIC	-268.219
Time:	05:24:56	BIC	-253.475
Sample:	0	HQIC	-262.227
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0008	0.008	0.105	0.916	-0.015	0.017
ma.L1	0.3078	0.076	4.028	0.000	0.158	0.457
ma.L2	-0.0936	0.095	-0.990	0.322	-0.279	0.092
ma.L3	-0.2106	0.088	-2.384	0.017	-0.384	-0.037
sigma2	0.0081	0.001	7.814	0.000	0.006	0.010

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	0.81
Prob(Q):	0.98	Prob(JB):	0.67
Heteroskedasticity (H):	1.09	Skew:	0.15
Prob(H) (two-sided):	0.76	Kurtosis:	2.77



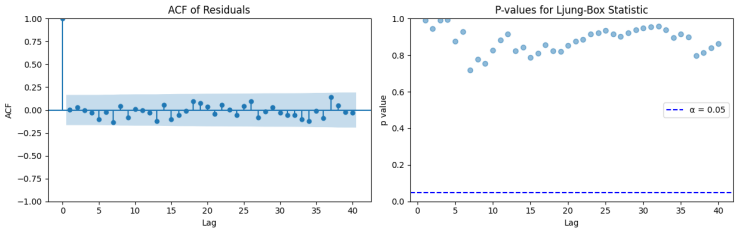


Figure 6: Statistical results for MA(3) and its residuals.

As seen in the images above, again, the residuals are normally distributed (slightly heavier on the tails), uncorrelated, and with p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant).

Comparing the 3 models calculated, the metrics obtained were:

Model	BIC	AIC	AICc
MA (1)	-1.827727	-1.890466	-1.889224
MA (2)	-1.792933	-1.876586	-1.874500
MA (3)	-1.797693	-1.902259	-1.899107

Table 2: Metrics for MA models

Again, taking into account the AIC metric, it can be confirmed that, out of the 3 models applied, the one that best fits our data is MA(3).
If we take into consideration that the models so far demonstrate both an autoregressive dynamic and a moving average shock, neither of the pure models can fully capture these components. Therefore, in the light of this knowledge, a combination of AR and MA terms will be explored, leading us to ARMA models.

4.3 ARMA Models

Considering that neither the AR nor the MA models are the best for the dataset being analysed, an ARMA model will be explored using the best orders obtained for p and q, so ARMA(2,3).

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(2, 0, 3)	Log Likelihood	143.425
Date:	Mon, 22 Dec 2025	AIC	-272.849
Time:	05:24:57	BIC	-252.208
Sample:	0	HQIC	-264.461
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0014	0.002	0.601	0.548	-0.003	0.006
ar.L1	-0.0821	0.113	-0.729	0.466	-0.303	0.139
ar.L2	0.7845	0.112	7.021	0.000	0.565	1.003
ma.L1	0.3669	0.140	2.612	0.009	0.092	0.642
ma.L2	-0.8891	0.138	-6.435	0.000	-1.160	-0.618
ma.L3	-0.4461	0.085	-5.235	0.000	-0.613	-0.279
sigma2	0.0077	0.001	6.650	0.000	0.005	0.010

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	0.52
Prob(Q):	0.95	Prob(JB):	0.77
Heteroskedasticity (H):	1.09	Skew:	0.09
Prob(H) (two-sided):	0.78	Kurtosis:	2.76

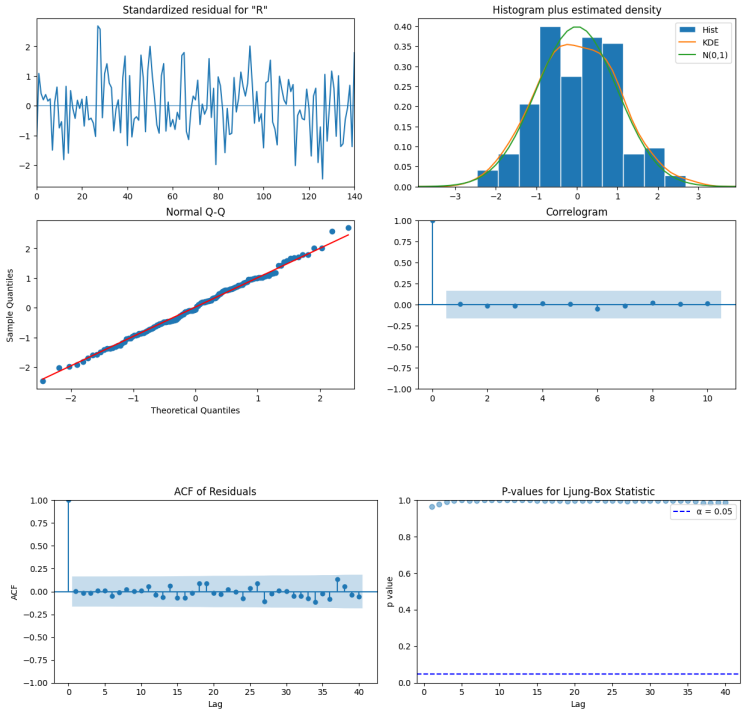


Figure 7: Statistical results for ARMA(2,3) and its residuals.

In this model, we can also see uncorrelated residuals, normally distributed, with p-values for the Ljung-Box test above 0.05.

Notice that all the MA terms (ma.L1, ma.L2, ma.L3) are highly significant (P>|z|= 0.000), while the AR terms show mixed significance. This suggests the data has a strong moving average component that the pure AR(2) model could not capture adequately.
To confirm that ARMA(2,3) was the best fit, combinations of these orders were implemented, and the AIC results are as follows:

Model	AIC
ARMA (1,3)	-1.915267
ARMA (2,3)	-1.935102
ARMA (3,3)	-1.911520
ARMA (2,1)	-1.916960
ARMA (2,2)	-1.870173

Table 3: AIC metric for ARMA models

Of all the ARMA models, it is clear that the one with the best AIC was indeed ARMA(2,3) so it can be considered that the best model for the dataset has been found.
In conclusion, all three models (AR, MA, and ARMA) pass the key diagnostic tests reasonably well, but the ARMA(2,3) combines both AR and MA components, making it more flexible in capturing the autocorrelation structure in the data.

5 Forecast

This section presents the forecasting stage of the analysis. Using the training–test split defined previously, we generate out-of-sample forecasts and assess their performance on the test period. To provide a meaningful point of comparison, we start with simple baseline methods (Naïve, Drift and Holt), which capture persistence and trend with minimal modelling assumptions. We then analyse an ARIMA$(2, 1, 3)$ specification, chosen based on the differencing requirement and the short-run dependence patterns identified earlier. For each approach, we report point forecasts and 95% prediction intervals, and we evaluate accuracy and interval calibration on the test set.

5.1 Baseline forecasts

Before analysing an ARIMA specification, it is important to define reference methods that capture what can already be achieved with minimal assumptions. These baselines serve two goals: (i) they provide an interpretable standard of comparison, and (ii) they prevent overestimating the value of a more complex model when the series is already highly persistent.

The following benchmark methods were considered:

- **Naïve forecast:** assumes the next value repeats the last observation,

$$\hat{y}_t = y_{t-1}.$$

- **Drift method:** The Drift forecast extrapolates the average change across the available training window. Let the training sample at time t be $\{y_1, \ldots, y_n\}$ (where y_n is the last value before t). The 1-step-ahead Drift forecast is

$$\hat{y}_{n+1} = y_n + \frac{y_n - y_1}{n-1}.$$

  In our implementation, 95% prediction intervals for Drift are constructed *approximately* by estimating the standard deviation s of the 1-step residuals

$$e_i = y_i - \left(y_{i-1} + \frac{y_n - y_1}{n-1} \right), \qquad i = 2, \ldots, n.$$

  and then using

$$[L_{n+1}, U_{n+1}] = \hat{y}_{n+1} \pm 1.96\, s,$$

  with L representing the Lower Bound and U the Upper Bound.

  This provides a simple uncertainty band based on empirical residual variability.

- **Holt's linear method:** Holt's method extends exponential smoothing by modelling both a level component ℓ_t and a trend component b_t:

$$\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}), \qquad b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta) b_{t-1},$$

  with the 1-step-ahead forecast

$$\hat{y}_{t+1|t} = \ell_t + b_t.$$

  In our implementation, the smoothing parameters (α, β) are estimated by optimisation on the current training window. For consistency with the other baselines, Holt 95% prediction intervals are also computed *approximately* using the in-sample residual standard deviation s from the fitted values:

$$[L_{t+1}, U_{t+1}] = \hat{y}_{t+1|t} \pm 1.96\, s.$$

The corresponding forecast plots for the baseline methods are provided in Appendix 9.5.

Note that the baseline methods use approximate 95% prediction intervals constructed from the empirical standard deviation of in-sample residuals, whereas ARIMA prediction intervals are obtained from the fitted statistical model.

5.2 ARIMA(2,1,3) forecast

After establishing baseline performance, we evaluate an ARIMA$(2, 1, 3)$ model, motivated by the need for first differencing and the short-run dependence structure identified earlier. Forecasts are produced using the same expanding-window 1-step-ahead protocol, and we report both point forecasts and 95% prediction intervals obtained from the fitted ARIMA model.

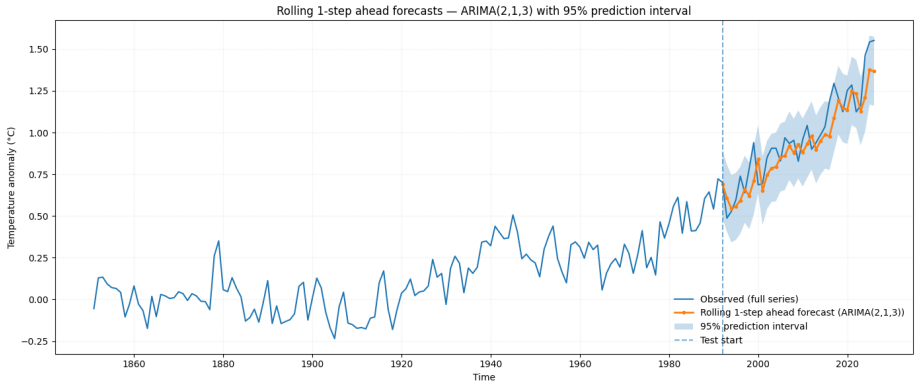


Figure 8: Forecast using Rolling 1-step ahead - ARIMA(2,1,3)

The detailed tables and the corresponding inside-count calculations are reported in the Appendix 9.6.

The "inside count" reports how many test observations fall within the corresponding 95% prediction intervals. In our rolling 1-step-ahead evaluation, forecasts were produced for 35 test years; in 31 of those years, the observed anomaly lays between the predicted lower and upper bounds. This corresponds to an empirical coverage of $31/35 \approx 0.886$, which is below the nominal 0.95 level and suggests that the ARIMA(2,1,3) prediction intervals are slightly too narrow (i.e., the model is overconfident) over the test period.

	Model	MAE	RMSE	95%_PI_Coverage
3	ARIMA(2,1,3)	0.0955	0.1172	0.886
1	Drift	0.0961	0.1191	0.943
0	Naïve	0.0980	0.1200	0.943
2	Holt	0.0981	0.1210	0.914

Figure 9: Rolling 1-step-ahead performance on the test period: baselines (Naïve, Drift, Holt) vs. ARIMA(2,1,3). Metrics: MAE, RMSE, and 95% PI coverage (approx. for baselines).

Figure 9 shows that ARIMA(2,1,3) achieves the lowest MAE and RMSE among

the methods considered, indicating slightly better point-forecast accuracy than the baselines. However, its empirical 95% prediction-interval coverage (0.886) is clearly below that of the baseline methods (approximately 0.914–0.943), meaning that its intervals are too narrow and the model is overconfident on the test period. Overall, ARIMA(2,1,3) is competitive for point predictions, but its uncertainty quantification is weaker than desired.

5.3 Exploring alternative ARIMA$(p, 1, q)$ specifications

Since the empirical 95% prediction-interval coverage obtained with ARIMA(2,1,3) was below the nominal level, we extended the analysis to a broader set of candidate models in order to find a better trade-off between point-forecast accuracy and interval quality. We tested all ARIMA$(p, 1, q)$ combinations with $p, q \in \{0, 1, 2, 3\}$ (16 models), which is consistent with the earlier dependence analysis (ACF/PACF) while avoiding overly parameterised specifications for an annual series.

All candidate models were evaluated using the same expanding-window, rolling 1-step-ahead procedure on the test period: for each test year t, the model was fitted using only observations up to $t - 1$, and a 1-step-ahead forecast and corresponding 95% prediction interval $[L_t, U_t]$ were produced.

For each ARIMA$(p, 1, q)$ model we computed:

- **MAE** and **RMSE**, which assess point-forecast accuracy;
- **95% interval coverage**[1] , defined as the proportion of test observations y_t satisfying $y_t \in [L_t, U_t]$;
- **Winkler score at 95%** [2](denoted Winkler_{95}), which evaluates interval forecasts by combining interval width with an explicit penalty when the observation falls outside the interval.

Let $\alpha = 0.05$ for a 95% interval. For each test year t, the Winkler interval score W_t is:

$$W_t = \begin{cases} (U_t - L_t), & \text{if } L_t \le y_t \le U_t, \\ (U_t - L_t) + \dfrac{2}{\alpha}(L_t - y_t), & \text{if } y_t < L_t, \\ (U_t - L_t) + \dfrac{2}{\alpha}(y_t - U_t), & \text{if } y_t > U_t. \end{cases}$$

[1]Interval calibration is assessed using empirical 95% coverage, i.e., the proportion of test observations that fall within the 95% prediction intervals. See more in reference [8].

[2]See references [6, 7] for details on interval evaluation and the Winkler score.

Thus, when the observation lies inside the interval, the score equals the interval width; otherwise, the score increases by a penalty proportional to how far the observation is outside the interval. The value reported in the table, Winkler_{95}, is the mean of W_t over all test years. Lower values indicate better interval forecasts.
Because Winkler_{95} accounts for both sharpness (narrow intervals) and calibration (penalising non-coverage), we rank the ARIMA$(p, 1, q)$ candidates by Winkler_{95} and report the 10 best models. Nevertheless, we also consider MAE and RMSE, as well as coverage, when interpreting the ranking. For example, as we can see in Figure 10, ARIMA$(3, 1, 3)$ and ARIMA$(3, 1, 2)$ remain competitive due to lower point errors while showing interval behaviour similar to the top-ranked candidates, motivating additional visual inspection.

	Model	MAE	RMSE	95%_Coverage	Avg_PI_Width	Winkler_95
0	ARIMA(2, 1, 0)	0.092052	0.112253	0.971429	0.427485	0.485983
1	ARIMA(3, 1, 0)	0.090208	0.115305	0.942857	0.413649	0.488685
2	ARIMA(2, 1, 1)	0.095029	0.116701	0.942857	0.412287	0.495254
3	ARIMA(3, 1, 1)	0.094291	0.116875	0.942857	0.412432	0.505453
4	ARIMA(2, 1, 3)	0.096453	0.118887	0.885714	0.409602	0.506094
5	ARIMA(2, 1, 2)	0.095374	0.117309	0.942857	0.412826	0.507271
6	ARIMA(3, 1, 3)	0.093810	0.116161	0.942857	0.406492	0.508476
7	ARIMA(3, 1, 2)	0.093227	0.115829	0.914286	0.405363	0.512942
8	ARIMA(0, 1, 2)	0.097375	0.118596	0.942857	0.416195	0.514917
9	ARIMA(1, 1, 2)	0.097745	0.119035	0.942857	0.416112	0.515374

Figure 10: Top 10 ARIMA$(p, 1, q)$ candidates (rolling 1-step ahead, 1991–2025), ranked by Winkler_{95}.

5.3.1 Shortlist and visual comparison of candidate ARIMA models

While the Top-10 table summarises performance numerically, visual inspection is essential to validate whether the apparent differences translate into meaningfully different forecast behaviour. In particular, plots allow us to check (i) whether candidate models produce consistent trajectories over the test period, (ii) whether prediction intervals are systematically wider/narrower, and (iii) whether the qualitative story changes when moving to a more complex specification.

Following the ranking in Figure 10, we first plot the four highest-ranked candidates (ARIMA(2,1,0), ARIMA(3,1,0), ARIMA(2,1,1), and ARIMA(3,1,1)), which all improve upon the initial ARIMA(2,1,3) in terms of interval quality (lower Winkler_{95}). We then add two additional models highlighted by their strong point-forecast accuracy, ARIMA(3,1,3) and ARIMA(3,1,2), even though they are not top-ranked by Winkler_{95}. The corresponding rolling 1-step-ahead plots are reported in Appendix 9.7.

Across the shortlist, the forecast trajectories are remarkably similar, with only minor differences in short-run fluctuations and interval width. The top-ranked ARIMA(2,1,0) provides a balanced compromise, achieving strong point accuracy while also delivering well-behaved intervals. ARIMA(3,1,0) attains the lowest MAE among the Top-10 candidates and remains very close to ARIMA(2,1,0) in terms of interval behaviour, suggesting that the main conclusions do not depend strongly on the exact AR/MA order. Moreover, increasing model complexity (e.g., moving to ARIMA(3,1,3)) does not materially change the forecast path, suggesting that the forecasts are driven primarily by the non-stationary level (captured by the $d = 1$ differencing), while the remaining serial dependence is weak. As a result, adding higher-order AR/MA terms provides only limited changes relative to simpler specifications.

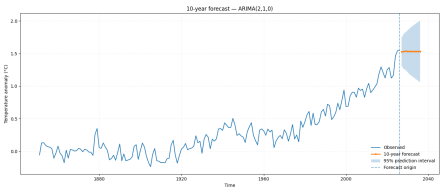
Overall, the fact that the best-performing models yield nearly identical trajectories is a robustness signal: the forecasts are relatively stable with respect to the choice of p and q within the tested range. At the same time, this stability limits the gains obtainable from more complex specifications, which helps explain why improvements over simple baselines are modest once the non-stationary level is accounted for (via $d = 1$) and the remaining short-run dependence is weak.

5.4 Ten-year-ahead forecasts

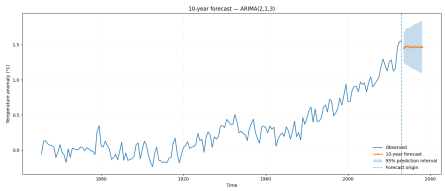
The rolling 1-step-ahead evaluation in Sections 5.1–5.3 assesses short-horizon performance under a realistic update cycle. We now complement that analysis with multi-step forecasts over a 10-year horizon, which are useful to illustrate how different modelling assumptions translate into qualitatively different long-run scenarios and uncertainty growth.

5.4.1 Forecast setup and models

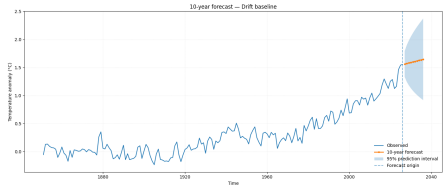
We produce 10-year-ahead forecasts from the end of the observed sample, reporting both point forecasts and 95% prediction intervals. We consider the best interval-ranked ARIMA model from Figure 10 (ARIMA(2,1,0)) and the initial candidate ARIMA(2,1,3), so that long-horizon behaviour can be compared under similar differencing but different short-run dynamics. In addition, we include the Drift baseline, since it was the most competitive benchmark in the test-period evaluation and represents an alternative assumption about long-run trend continuation.



(a) ARIMA(2,1,0)



(b) ARIMA(2,1,3)



(c) Drift

Figure 11: Ten-year-ahead forecasts (ARIMA(2,1,0), ARIMA(2,1,3) and Drift).

The three plots shown here, together with an additional 10-year-ahead forecast plot, are included in Appendix 9.8.

5.4.2 Interpretation of long-horizon behaviour

The two ARIMA specifications with $d = 1$ and no drift (ARIMA(2,1,0) and ARIMA(2,1,3)) tend to produce nearly flat forecasts as the horizon increases. Intuitively, first differencing models the year-to-year changes; as we move further into the future, the expected change is driven toward zero, so the forecast for the level converges to an approximately constant value. Differences between ARIMA(2,1,0) and ARIMA(2,1,3) are therefore most visible at short horizons: ARIMA(2,1,3) can display slightly more short-term oscillation because the additional MA component incorporates recent forecast errors (shocks), whereas ARIMA(2,1,0) typically stabilises faster.
In contrast, the Drift method conveys a qualitatively different message: by extrapolating an average historical slope, it continues to increase over the horizon. This highlights an important point for interpretation: the warming signal is clearly present in the historical data, but the shape of a 10-year forecast depends on whether the model encodes a deterministic trend component (as Drift does) or whether long-run changes are expected to die out (as in a differenced ARIMA without drift).

5.4.3 Uncertainty over the 10-year horizon

Across all methods, the 95% prediction intervals widen with the forecast horizon, reflecting the accumulation of uncertainty when predicting multiple steps ahead. This widening is expected and should be emphasised when presenting long-horizon results: the point forecast alone is not sufficient, and the interval

expansion communicates that uncertainty grows rapidly as we forecast further into the future.

5.4.4 Discussion and link to model choice

Taken together, the 10-year-ahead plots support two conclusions. First, the long-horizon trajectory is relatively stable across the best-performing ARIMA candidates, reinforcing the robustness observed in the shortlist comparison. Second, the Drift baseline provides a useful alternative scenario because it explicitly propagates a long-run slope. For this reason, the long-horizon results are best interpreted as scenario-dependent: ARIMA without drift yields a conservative "plateau" behaviour, whereas Drift yields continued increases. This distinction should be kept in mind when drawing conclusions about long-term warming from purely statistical extrapolation.

6 Results and Validation

This section consolidates the key forecasting results and validates the final conclusions using out-of-sample evaluation on the test period, together with interval-based performance diagnostics.

6.1 Test-period results (rolling 1-step ahead)

Forecast performance was assessed on the test period (1991–2025) using an expanding-window, rolling 1-step-ahead protocol, ensuring that each prediction is strictly out-of-sample. The baseline methods (Naïve, Drift, and Holt) provide an essential reference for what can be achieved with minimal assumptions. As shown in Figure 9, ARIMA(2,1,3) attains slightly better point accuracy (lowest MAE and RMSE), but its empirical 95% prediction-interval coverage is clearly below the nominal level, indicating underestimated uncertainty (overconfident intervals). This result motivates moving beyond a single ARIMA specification and evaluating alternative ARIMA$(p, 1, q)$ candidates under the same test protocol.

6.2 Validation of interval quality and model choice

To compare ARIMA candidates fairly, we evaluate both point accuracy (MAE/RMSE) and interval behaviour. Figure 10 summarises the ARIMA$(p, 1, q)$ grid with $p, q \in \{0, 1, 2, 3\}$, reporting coverage and the Winkler$_{95}$ score. Ranking by Winkler$_{95}$ prioritises models that achieve a better trade-off between sharpness (narrower intervals) and calibration (penalising non-coverage), addressing the main weakness observed for ARIMA(2,1,3). The shortlist plots in Appendix 9.7 further support the table results: top-ranked candidates (e.g., ARIMA(2,1,0) and ARIMA(3,1,0)) deliver comparable trajectories and interval behaviour, while models highlighted by low MAE/RMSE (ARIMA(3,1,3) and

ARIMA(3,1,2)) remain competitive but do not materially change the forecast narrative.
Taken together, these results validate two practical points: (i) conclusions are not sensitive to the exact choice of p and q within the tested range, because the best candidates produce very similar forecast paths; and (ii) interval calibration must be considered explicitly, since selecting models purely by MAE/RMSE can favour specifications that are overconfident on the test period.

6.3 Long-horizon validation (10-year-ahead forecasts)

The 10-year-ahead forecasts (Figure 11) are included to illustrate how model assumptions affect long-run behaviour and uncertainty growth. The ARIMA models without drift (ARIMA(2,1,0) and ARIMA(2,1,3)) tend to stabilise over longer horizons, whereas the Drift baseline continues to increase by construction. The prediction intervals widen substantially with the horizon for all methods, emphasising that long-horizon projections should be interpreted together with uncertainty bands. An additional long-horizon forecast for another shortlisted ARIMA model is provided in Appendix 9.8.1 to confirm that the qualitative conclusions remain robust across alternative (p, q) choices.

Overall, the validation results above motivate the discussion in the next section, where we summarise the main benefits and limitations of the modelling choices and interpret the practical implications of the forecasts and their associated uncertainty.

7 Benefits and Limitations

This section summarises the main strengths and weaknesses of the forecasting approach used in this project, focusing on what the methods allow us to conclude and what should be interpreted with care.

7.1 Benefits

- **Realistic evaluation protocol.** The expanding-window, rolling 1-step-ahead procedure assesses performance in a setting that closely matches practical use, where the model is updated as each new year becomes available.
- **Meaningful benchmarks.** The Naïve, Drift, and Holt baselines provide a clear reference for what can be achieved with minimal assumptions, helping avoid overvaluing more complex models.
- **Temporal structure modelling.** By using $d = 1$ differencing, the model focuses on year-to-year changes rather than raw levels. The AR and MA terms (p and q) then capture any remaining short-run dependence in these

changes (autocorrelation), providing a structured way to model the series' "memory" beyond simple trend extrapolation.

- **Uncertainty communication.** ARIMA produces model-based 95% prediction intervals, which are essential for communicating uncertainty rather than relying only on point forecasts.
- **Robust qualitative conclusions.** Several well-performing ARIMA candidates yield very similar trajectories, suggesting that the main forecasting message is stable within the tested (p, q) range.

7.2 Limitations

- **Limited gains over simple methods.** Because the series is dominated by a strong non-stationary component, simple baselines remain highly competitive, and improvements in MAE/RMSE from ARIMA are modest.
- **Long-horizon behaviour depends on trend assumptions.** Ten-year-ahead projections are scenario-dependent: differenced ARIMA models without drift tend to stabilise, while Drift continues to increase by construction. Consequently, long-horizon forecasts should be interpreted as assumption-driven scenarios rather than a single definitive trajectory.
- **Intervals may be miscalibrated.** Some models exhibit empirical 95% coverage below the nominal level, implying overconfident prediction intervals. This shows that model selection cannot rely on MAE/RMSE alone and must include interval diagnostics.
- **Risk of overfitting with higher orders.** Increasing p and q can slightly improve point accuracy in some cases, but added complexity may not generalise well, particularly with annual data and a limited test window.

Closing remark: Overall, the rolling 1-step-ahead evaluation indicates that ARIMA models can yield a modest improvement in point-forecast accuracy relative to simple benchmarks, but interval calibration (95% coverage and Winkler_{95}) provides an important complement to MAE/RMSE when selecting a final specification. The comparison across ARIMA$(p, 1, q)$ candidates suggests that the main qualitative conclusions are robust within the tested model family. At longer horizons, however, forecasts are scenario-dependent: models that do not propagate a deterministic trend tend to stabilise, whereas the Drift baseline continues to increase by construction. For this reason, long-horizon results should be interpreted as assumption-driven scenarios and reported together with the widening prediction intervals.

8 References

References

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[4] Cleveland, R. B., Cleveland, W. S., McRae, J. E., and Terpenning, I. (1990). *STL: A seasonal-trend decomposition procedure based on loess*. Journal of Official Statistics, 6(1), 3–73.

[5] Crawley, Michael J. (2007) *The R Book*, Chapter 9 "Statistical Modelling", Section "The Principle of Parsimony (Occam's Razor)" Available at: `https://www.oreilly.com/library/view/the-r-book/9780470510247/ch009-sec004.html`

[6] Hyndman, R. J. (2018). *Forecast Evaluation* (ACEMS Forecasting Workshop slides). Includes prediction-interval evaluation and the Winkler interval score. Available at: `https://robjhyndman.com/acemsforecasting2018/1-Forecast-Evaluation.pdf`

[7] Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts (online textbook). See chapters on forecast accuracy and prediction intervals. Available at: `https://otexts.com/fpp3/`

[8] Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.), Section "Prediction intervals". (Defines prediction intervals and their coverage probability, including 95% intervals.) Available at: `https://otexts.com/fpp3/prediction-intervals.html`

9 Appendix

9.1 Model AR(1)

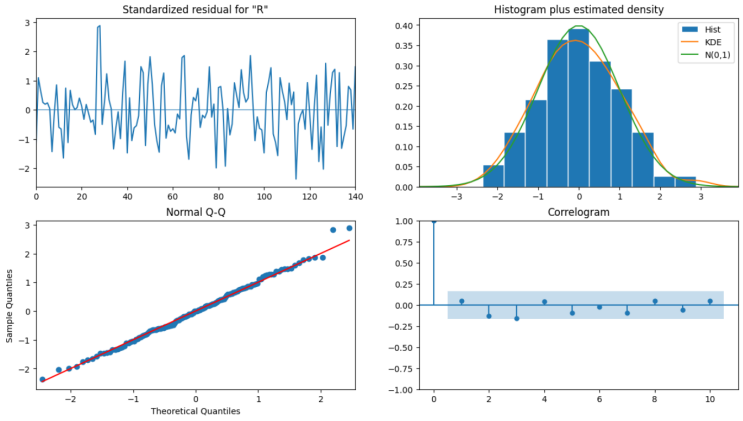
The results obtained for the model AR(1) are the following:

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(1, 0, 0)	Log Likelihood	135.188
Date:	Sun, 21 Dec 2025	AIC	-264.375
Time:	23:22:20	BIC	-255.529
Sample:	0	HQIC	-260.780
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0010	0.011	0.093	0.926	-0.020	0.022
ar.L1	0.2712	0.076	3.583	0.000	0.123	0.420
sigma2	0.0086	0.001	7.918	0.000	0.006	0.011

Ljung-Box (L1) (Q):	0.30	Jarque-Bera (JB):	0.85
Prob(Q):	0.58	Prob(JB):	0.65
Heteroskedasticity (H):	1.20	Skew:	0.17
Prob(H) (two-sided):	0.53	Kurtosis:	2.85



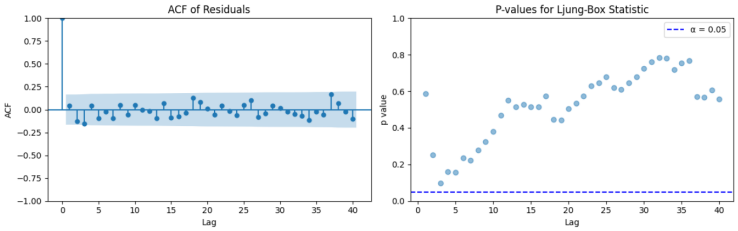


Figure 12: Statistical results for AR(1) and its residuals.

As seen in the images above, the residuals are normally distributed, uncorrelated, and have p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant).

9.2 Model AR(3)

The results obtained for the model AR(3) are the following:

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(3, 0, 0)	Log Likelihood	138.133
Date:	Sun, 21 Dec 2025	AIC	-266.266
Time:	23:22:22	BIC	-251.522
Sample:	0	HQIC	-260.274
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0008	0.009	0.098	0.922	-0.016	0.017
ar.L1	0.3032	0.078	3.884	0.000	0.150	0.456
ar.L2	-0.1440	0.091	-1.576	0.115	-0.323	0.035
ar.L3	-0.1020	0.081	-1.265	0.206	-0.260	0.056
sigma2	0.0082	0.001	7.791	0.000	0.006	0.010

Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	1.00
Prob(Q):	1.00	Prob(JB):	0.61
Heteroskedasticity (H):	1.17	Skew:	0.17
Prob(H) (two-sided):	0.60	Kurtosis:	2.77

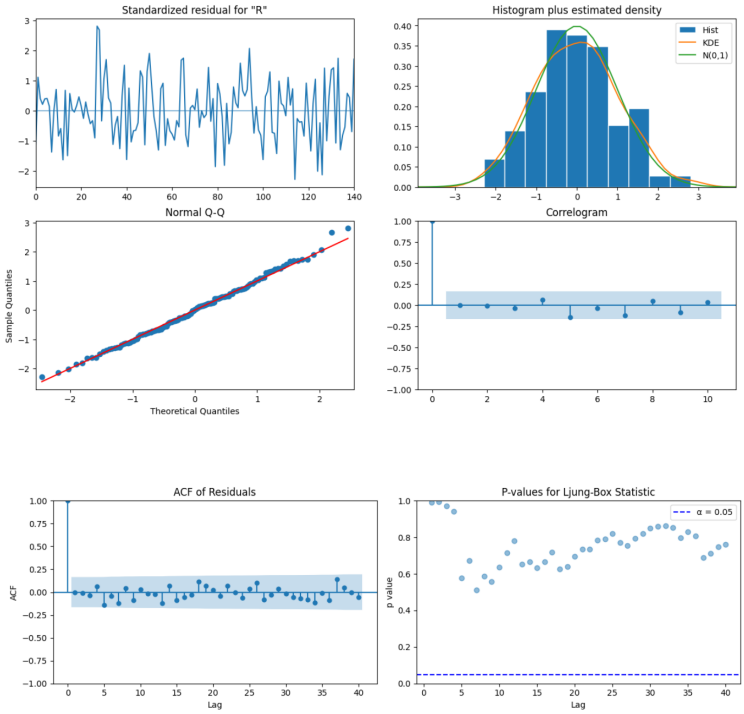


Figure 13: Statistical results for AR(3) and its residuals.

As seen in the images above, the residuals are normally distributed, uncorrelated, and have p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant).

9.3 Model MA(1)

The results obtained for the model MA(1) are the following:

```
                               SARIMAX Results
==============================================================================
Dep. Variable:              Remainder   No. Observations:                  141
Model:                 ARIMA(0, 0, 1)   Log Likelihood                 136.278
Date:                Mon, 22 Dec 2025   AIC                           -266.556
Time:                        05:24:54   BIC                           -257.709
Sample:                             0   HQIC                          -262.961
                                  - 141
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0010      0.010      0.099      0.921      -0.019       0.021
ma.L1          0.3071      0.072      4.241      0.000       0.165       0.449
sigma2         0.0085      0.001      7.937      0.000       0.006       0.011
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 0.51
Prob(Q):                              0.98   Prob(JB):                         0.77
Heteroskedasticity (H):               1.23   Skew:                             0.12
Prob(H) (two-sided):                  0.47   Kurtosis:                         2.82
===================================================================================
```

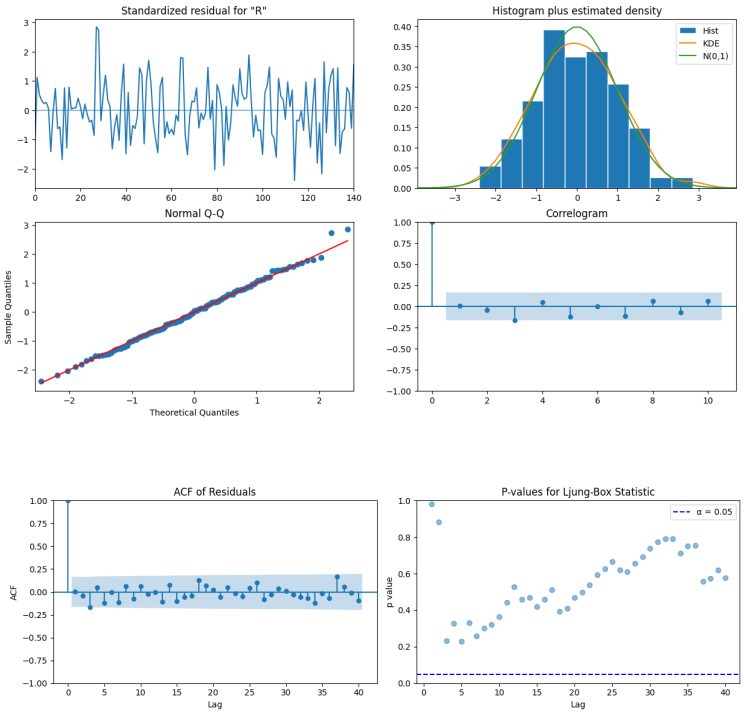


Figure 14: Statistical results for MA(1) and its residuals.

As seen in the images above, although the residuals are normally distributed, uncorrelated, and with p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant), not all the model's coefficients support the null hypothesis: the constant presents a p-value of 0.921 (>0.05).

9.4 Model MA(2)

The results obtained for the model MA(2) are the following:

SARIMAX Results

Dep. Variable:	Remainder	No. Observations:	141
Model:	ARIMA(0, 0, 2)	Log Likelihood	136.299
Date:	Mon, 22 Dec 2025	AIC	-264.599
Time:	05:24:55	BIC	-252.804
Sample:	0	HQIC	-259.806
	- 141		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0010	0.011	0.095	0.925	-0.020	0.022
ma.L1	0.3192	0.076	4.174	0.000	0.169	0.469
ma.L2	0.0190	0.094	0.201	0.841	-0.166	0.204
sigma2	0.0085	0.001	7.899	0.000	0.006	0.011

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	0.55
Prob(Q):	0.94	Prob(JB):	0.76
Heteroskedasticity (H):	1.23	Skew:	0.12
Prob(H) (two-sided):	0.48	Kurtosis:	2.81

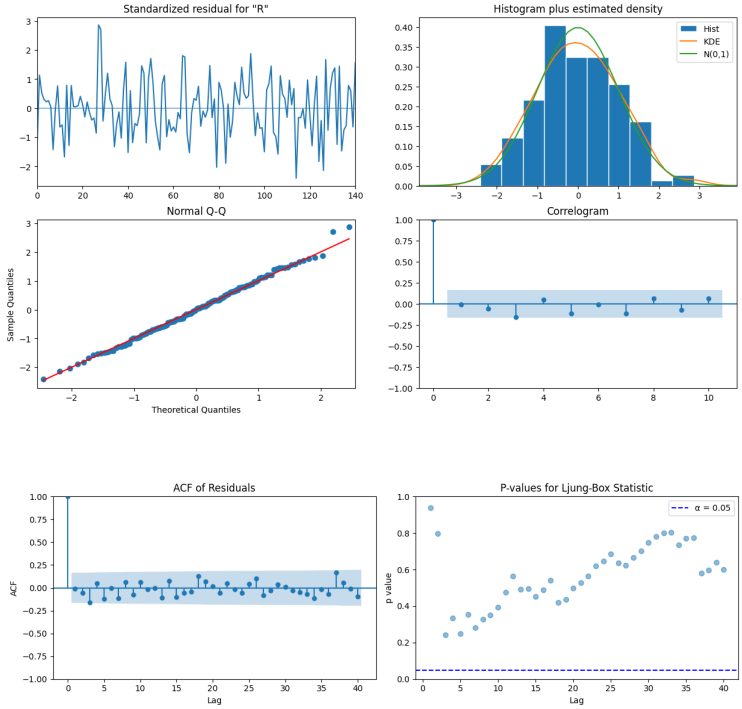


Figure 15: Statistical results for MA(2) and its residuals.

As seen in the images above, although the residuals are normally distributed, uncorrelated, and with p-values for the Ljung-Box test above 0.05 (meaning they are statistically significant), not all the model's coefficients support the null hypothesis: the constant presents a p-value of 0.921 while the second coefficient has a value P>|z|of 0.841 (>0.05).

9.5 Baseline forecast plots

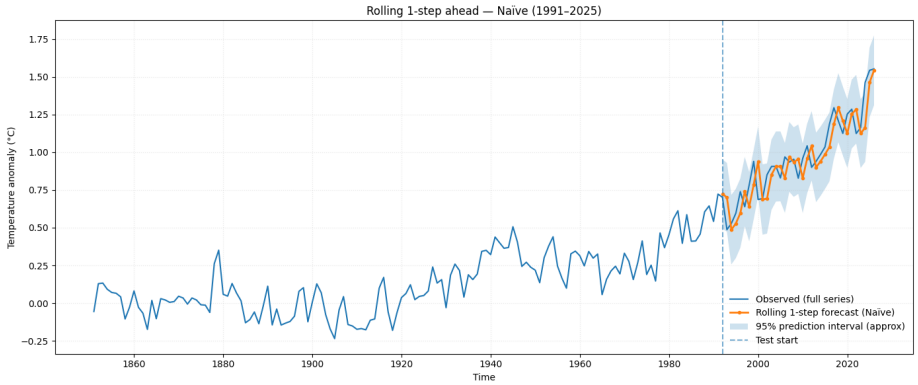


Figure 16: Forecast using Rolling 1-step ahead - Naïve

Model	MAE	RMSE
Naïve	0.0980	0.1200

Table 4: Naïve baseline performance (rolling 1-step ahead, 1991–2025)

Metric	Value
Approx. 95% Coverage	0.943

Table 5: Naïve baseline interval diagnostic (rolling 1-step ahead, 1991–2025)

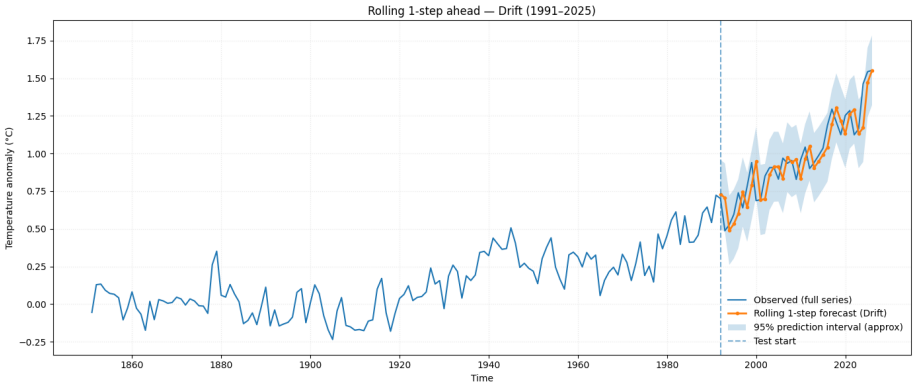


Figure 17: Forecast using Rolling 1-step ahead - Drift

Model	MAE	RMSE
Drift	0.0961	0.1191

Table 6: Drift baseline performance metrics (rolling 1-step ahead, 1991–2025)

Metric	Value
Approx. 95% Coverage	0.943

Table 7: Drift baseline interval diagnostic (rolling 1-step ahead, 1991–2025)

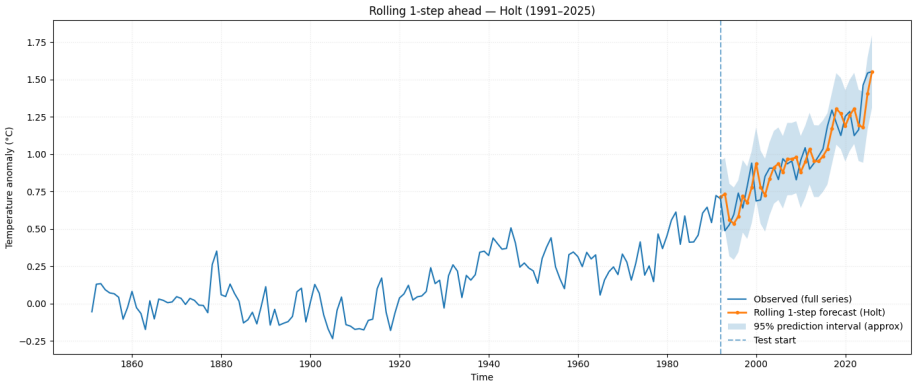


Figure 18: Forecast using Rolling 1-step ahead - Holt

Model	MAE	RMSE
Holt	0.0981	0.1210

Table 8: Holt baseline performance metrics (rolling 1-step ahead, 1991–2025)

Metric	Value
Approx. 95% Coverage	0.914

Table 9: Holt baseline interval diagnostic (rolling 1-step ahead, 1991–2025)

9.6 ARIMA(2,1,3) Results

Model	MAE	RMSE
ARIMA(2,1,3)	0.0955	0.1172

Table 10: ARIMA(2,1,3) performance metrics (rolling 1-step ahead, 1991–2025)

Metric	Value
95% PI Coverage	0.8857
Inside count (points within PI)	31 / 35

Table 11: ARIMA(2,1,3) prediction interval diagnostics (rolling 1-step ahead, 1991–2025)

We report the inside count explicitly for ARIMA(2,1,3) to make the coverage calculation transparent for the model-based 95% prediction intervals. For the baseline methods, prediction intervals are constructed using a residual-based approximation, so we summarize their uncertainty performance using coverage values only.

9.7 Shortlist and visual comparison of candidate ARIMA models

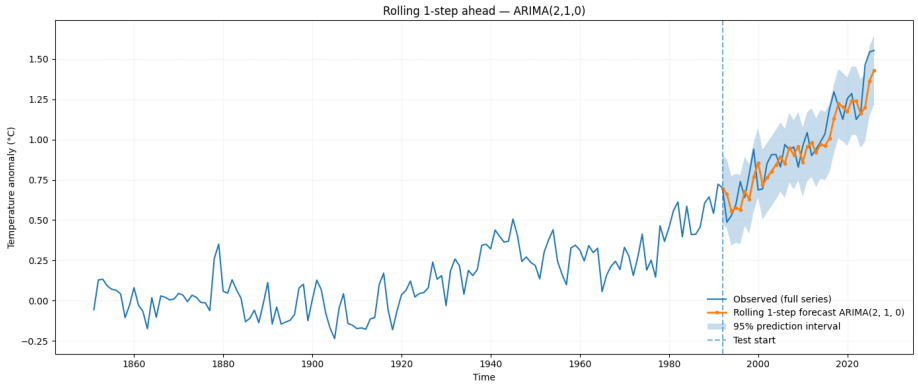


Figure 19: Forecast using Rolling 1-step ahead - ARIMA(2,1,0)

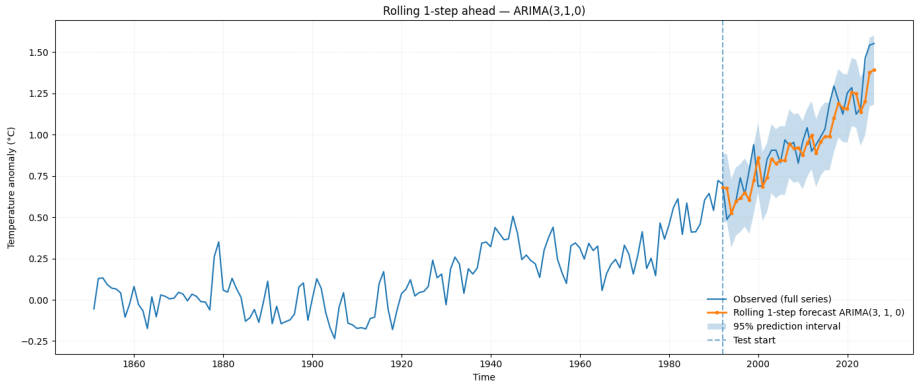


Figure 20: Forecast using Rolling 1-step ahead - ARIMA(3,1,0)

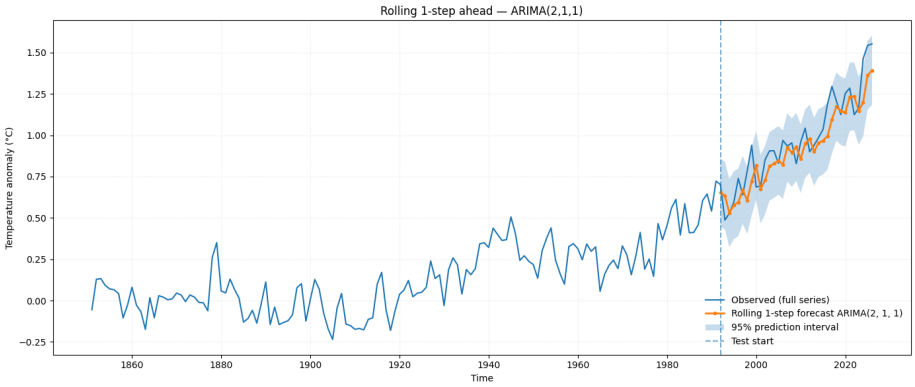


Figure 21: Forecast using Rolling 1-step ahead - ARIMA(2,1,1)

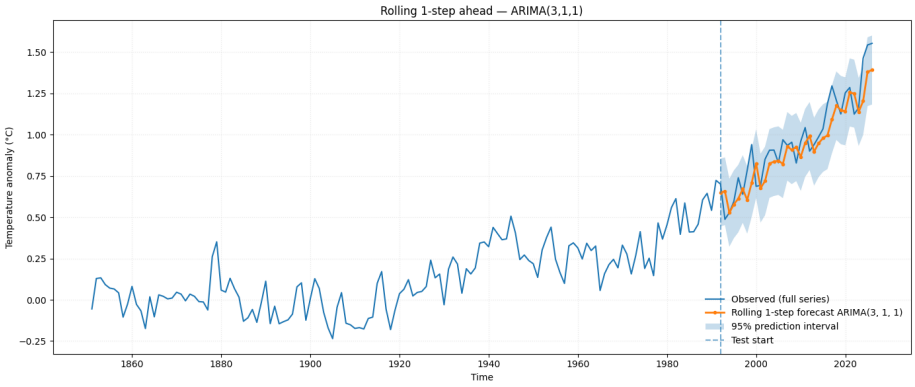


Figure 22: Forecast using Rolling 1-step ahead - ARIMA(3,1,1)

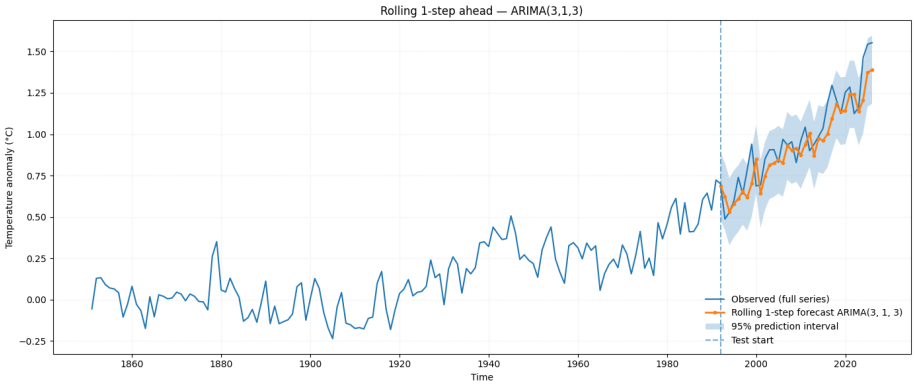


Figure 23: Forecast using Rolling 1-step ahead - ARIMA(3,1,3)

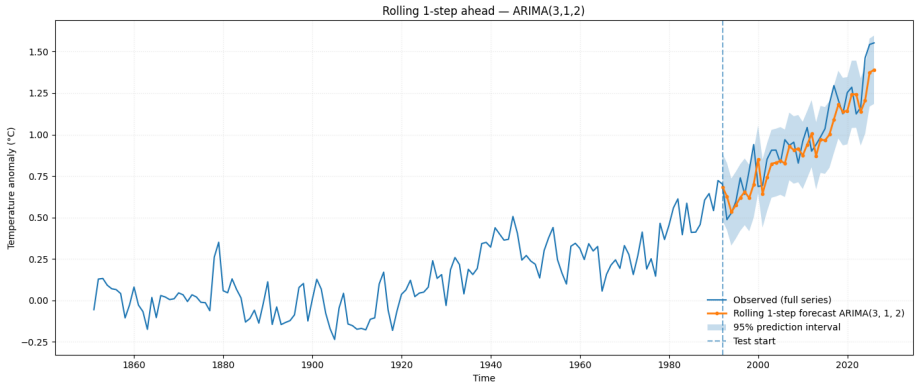


Figure 24: Forecast using Rolling 1-step ahead - ARIMA(3,1,2)

9.8 10-year-ahead forecast plots

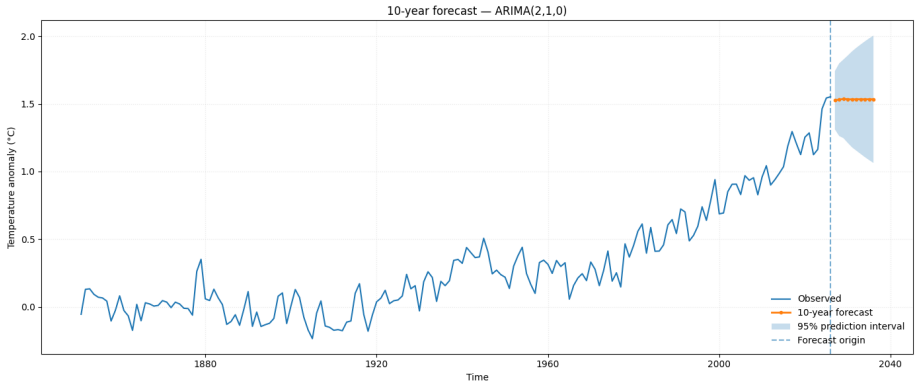


Figure 25: 10-year ahead forecast - ARIMA(2,1,0)

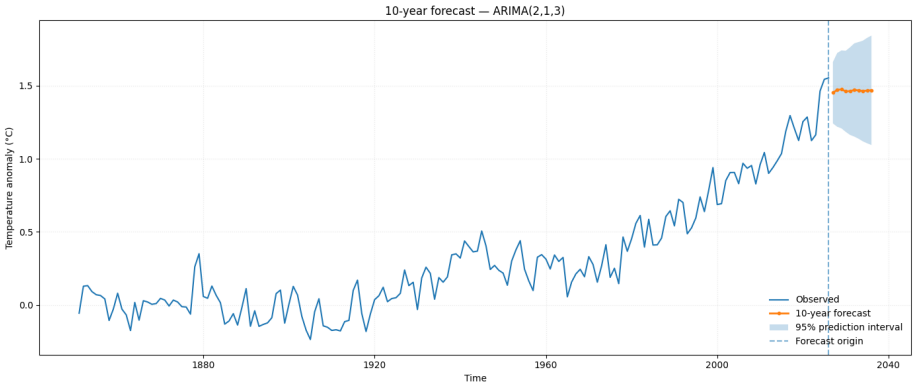


Figure 26: 10-year ahead forecast - ARIMA(2,1,3)

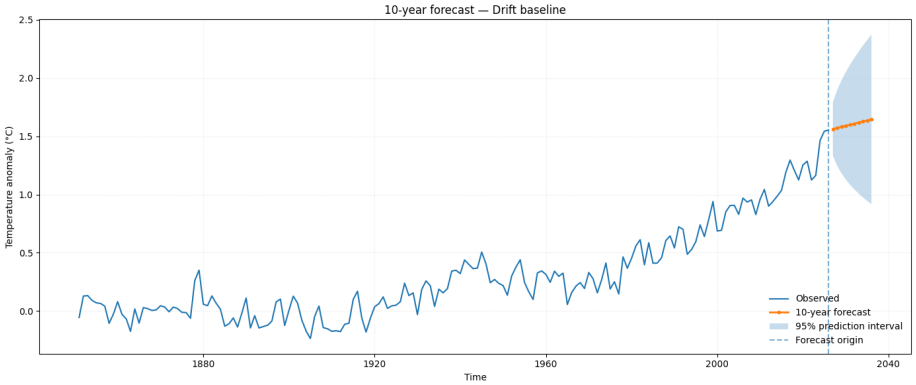


Figure 27: 10-year ahead forecast - Drift

9.8.1 10-year-ahead forecast additional plot

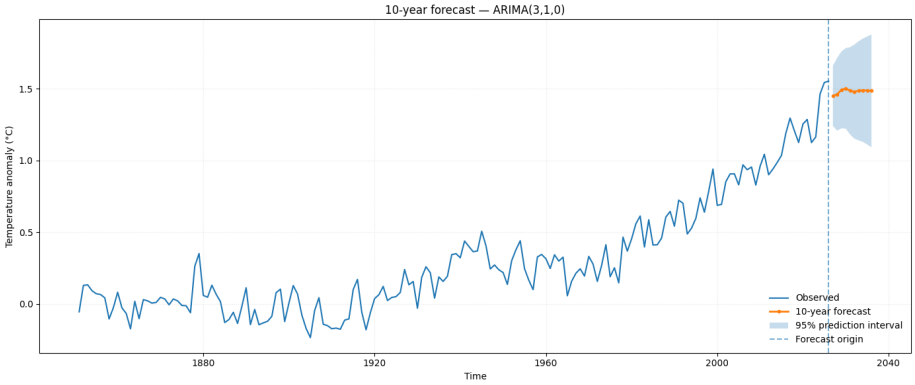


Figure 28: 10-year ahead forecast - ARIMA(3,1,0)

An additional long-horizon forecast for the second best ARIMA candidate is shown here to verify that the qualitative conclusions (plateau vs. drift and widening uncertainty) are robust to the choice of (p, q) within the tested range.